A VALUATION RING IS NOETHERIAN EXACTLY WHEN IT IS A DISCRETE VALUATION RING OR A FIELD

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Editorial note. The counted proof refers to the correspondence between ideals and ideals of the value group by label; it is recorded as a navigation stub pointed at its \label line, and the argument consuming it stays inside the counted span.

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Allowed standard prerequisites (editorial).

- Ideals of a valuation ring correspond to ideals of the value group. Used in both halves of the counted proof (lines 12323 and 12330). Translates the ideal-theoretic chain conditions into statements about the value group.
- The value group of a discrete valuation ring is the integers. Used in the first half of the counted proof (lines 12320-12325). Produces explicit generators of the ideals and hence a principal ideal domain.

Source reference. Original lemma 00IH: sources/stacks-algebra/pinned-original-source.tex, source lines 12302–12302. Navigation pointer for \ref{lemma-ideals-valuation-ring} used inside the counted span (line 12323); the argument that consumes it is retained in the span.

Lemma 0.1. *A valuation ring is Noetherian if and only if it is a discrete valuation ring or a field.*

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Proof. Suppose A is a discrete valuation ring with valuation $v : A \setminus \{0\} \to \mathbf{Z}$ normalized so that $\mathrm{Im}(v) = \mathbf{Z}_{\geq 0}$. By Lemma 00IH the ideals of A are the subsets $I_n = \{0\} \cup v^{-1}(\mathbf{Z}_{\geq n})$. It is clear that any element $x \in A$ with $v(x) = n$ generates I_n. Hence A is a PID so certainly Noetherian.

Suppose A is a Noetherian valuation ring with value group Γ. By Lemma 00IH we see the ascending chain condition holds for ideals in Γ. We may assume A is not a field, i.e., there is a $\gamma \in \Gamma$ with $\gamma > 0$. Applying the ascending chain condition to the subsets $\gamma + \Gamma_{\geq 0}$ with $\gamma > 0$ we see there exists a smallest element γ_0 which is bigger than 0. Let $\gamma \in \Gamma$ be an element $\gamma > 0$. Consider the sequence of elements γ, $\gamma - \gamma_0$, $\gamma - 2\gamma_0$, etc. By the ascending chain condition these cannot all be > 0. Let $\gamma - n\gamma_0$ be the last one ≥ 0. By minimality of γ_0 we see that $0 = \gamma - n\gamma_0$. Hence Γ is a cyclic group as desired. $\square$